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Abdelhak Bourdim

Senior Embedded Software Engineer — C/C++ • RTOS • Embedded Linux

Date of Birth: August 20, 1969

Nationality: French

Status: Married

Driving License: B • Eligible for Luxembourg cross-border permit

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Location: Paris region, France

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Website: workshop-diy.org

Mobility: Luxembourg (cross-border from France) — Available immediately

PROFILE

Embedded software specialist with **over 20 years of experience** developing firmware, middleware and application layers — bare metal, RTOS and Embedded Linux — for safety-critical and industrial systems. Design of low-level drivers, communication stacks (CAN, BLE, USB, TCP/IP) and embedded applications on ARM Cortex, STM32, Microchip and NXP architectures.

Certified solutions for **aerospace** (Safran, Thales — DO-178), **medical devices** (GE Healthcare, Sorin — IEC 62304), **automotive** (Valeo, Arkamys), **IoT cybersecurity** (Enedis — Linky), **payment systems** (Cartadis) and **smart EV charging V2G** (E2-CAD — ISO 15118). Trilingual French/English/Arabic, a strong asset for Luxembourg's multilingual market.

PROFESSIONAL EXPERIENCE

E2-CAD — Eragny, France

Embedded Software Engineer — EV Charging / V2G

EV Charging Station — Iotecha IP

- Reverse engineering and source code analysis (static and dynamic architecture)
- Embedded code review on 3 boards (Microchip PIC, STM32), debug tools (ST-Link, nsprog)
- Communication protocol analysis between Linux modules and embedded boards (RS232, I2C, Saleae)
- Python extension development and real-time HTML web report generation
- Test bench design and setup: Chargebyte tools integration, EV simulator
- Debug and commissioning of charging stations (PLC, HomePlug Green PHY, ISO 15118, OCPP)

Env: Embedded Linux, C/C++, Microchip PIC, STM32, PLC, HomePlug Green PHY, ISO 15118, OCPP, CAN, TCP/IP, Saleae, HackRF One, Python, Bash

Safran — Creteil, France

Embedded Software Engineer — Aerospace (SSPC)

Solid State Power Controller — Low-level Drivers

- Development and update of low-level C drivers for the SSPC system
- Logging tools setup for diagnostics and exchange traceability
- Writing and executing on-target validation tests with Lauterbach TRACE32
- Bash/Awk scripting for test process automation

Env: C, Lauterbach TRACE32, Bash, Awk

Asterios Technologies — Massy, France

Embedded Software Engineer — Aerospace (SOM Tests)

- C validation test development for SOM module (T1042 processor)
- Processor errata analysis and software workaround identification
- Test documentation writing per project requirements

Env: C, Bash, Python, Lauterbach TRACE32, Polarion

Thales — Gennevilliers, France

Embedded Software Engineer — Defense Radio

LPM + N-MMR Projects — Mission Preparation & Multi Mode Receiver

- Application layer development under Embedded Linux in C/C++ for 2 projects
- Configuration management and automation scripts (Bash, Curl, Tshark)
- HLT (High Level Tests) development and execution
- Bug fixing and network protocol analysis (Websockets)

Env: Embedded Linux, C/C++, Bash, Websockets, Curl, Tshark, Python

Valeo — Creteil, France / Egypt

Embedded Software Engineer — Test Audit

- Audit of test processes and tools (NI Mastre, Vector CANoe)
- Automation proposals and client/system requirements review

Env: NI Mastre, Vector CANoe

Arkamys — Levallois-Perret, France

Embedded Software Engineer — Automotive Audio

- GStreamer plugin and CAN driver development (ELM327), diagnostic tools
- Unit & integration testing (Valgrind), Buildroot build system

Env: Embedded Linux, C/C++, Buildroot, GStreamer, Valgrind, CAN, Python, Jira

PROFESSIONAL EXPERIENCE

Thales — Velizy-Villacoublay, France

Nov 2020 – Jun 2021

Embedded Software Engineer — Autonomous Train (ATO)

- 32-bit to 64-bit application migration on Embedded Linux platform
- IPC communication protocol definition and implementation
- Network diagnostic tools and integration test development

Env: Embedded Linux, C, UDP, TCP, Wireshark, Tshark, Python, Jira

PK Vitality — Paris, France

Jun – Sep 2020

Embedded Software Engineer — Medical Device (CGM)

- HAL development (AFE AD5940, PMIC DA9070), QSPI driver, BLE NRF52840
- STM32 touchscreen TouchGFX, FreeRTOS, low power mode
- In vitro testing and Doxygen documentation (IEC 62304 compliant)

Env: STM32L459, NRF52840, C, C++, FreeRTOS, BLE, TouchGFX, Doxygen

Zodiac Aerospace (Safran) — Montreuil / Vancouver, Canada

Feb 2017 – Mar 2020

Embedded Software Engineer — Aerospace (SSPC A350)

- Low-level layer adaptation: CAN, RS485 drivers, thermal memory (dsPIC33)
- uAFDX to CAN gateway implementation and diagnostic functions
- SSPC A350 application layer development and optimization
- Validation testing and aerospace-standard documentation
- Flight test campaign at UBC Vancouver, Canada (DO-178 certification)

Env: Microchip dsPIC33, MPLAB, TI Concerto, CCS Studio, C

Enedis — Nanterre, France

May 2015 – Sep 2016

Embedded Cybersecurity Engineer — Smart Grid

- Security diagnostics and audit of Linky smart meter concentrators
- Penetration testing on Embedded Linux systems (Kali Linux, Nmap)
- Protocol sniffing and fuzzing (Wireshark, Tshark, Scapy, SDR)
- Code audit and review with manufacturers (Sonar, PC Lint)

Env: Kali Linux, Embedded Linux, Nmap, Wireshark, Scapy, SDR, Python, Bash

Sorin — Clamart, France

Nov 2014 – Apr 2015

Embedded Software Engineer — Medical (BLE Pacemaker)

- Bluetooth Smart module validation, Serial Port Service Profile
- Smartphone communication protocol and Android application

Env: Cortex M0, Keil, C, BLE, Android

General Electric Healthcare — Buc / USA

2012 – 2014

Embedded Software Engineer — Medical Imaging

- CANOpen stack integration (slave/master), CAN drivers and services
- CMX-RTX kernel porting, US team technical support

Env: TI TMS320, ARM Cortex A8, VxWorks, C, C++

Photomaton / BCM / Vertical M2M / PGS — Paris

2010 – 2012

Payment systems and remote metering — Full embedded + PC development on NXP LPC (USB, RFID, GSM/GPRS, Bluetooth)

Cartadis — Fontenay-sous-Bois, France

Oct 2001 – Dec 2009

8 years — Access control and payment terminals: drivers, embedded applications, USB/TCP/IP stacks, DES/RC4 encryption (NEC V853, Hitachi H8S, NXP LPC, ATMEL AT91)

TECHNICAL SKILLS

Languages	C, C++, Python, Bash, Assembly, Perl, Awk
OS / RTOS	Embedded Linux, VxWorks, FreeRTOS, UCOSII, Tnkernel
MCUs	ARM Cortex (M0, M4, A8), STM32, NRF52, ESP32, Microchip PIC/dsPIC, NXP LPC, TI DSP/Concerto
Protocols / Bus	CAN, CANOpen, SPI, I2C, RS232, RS485, UART, USB, BLE, AFDX, Ethernet, TCP/UDP
Stacks	TCP/IP, USB Host/Device/HID, BLE, CANOpen, RFID, GStreamer
Security / Payment	Smart Card, RFID Mifare, DES, RC4, USB CCID, embedded pentesting
Tools	Lauterbach TRACE32, IAR, Keil, MPLAB, CCS Studio, Green Hills, Buildroot, CubeMx, TouchGFX
Management	Git, Synergy, Jira, Polarion, Doxygen, Wireshark, Saleae

STANDARDS & METHODS

DO-178	MISRA C	IEC 62304	ISO 15118 (V2G)	ISO 14443 (Smart Card)	ISO 15693 (RFID)	OCPP	V-Model	Agile / Scrum
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EDUCATION

AFPA, Champs-sur-Marne, France

Engineer — Electronic and Computer Systems Design

INDUSTRY SECTORS

Aerospace (Safran, Thales, Zodiac)

Medical Devices (GE Healthcare, Sorin, PK Vitality)

Tenon Hospital, Paris, France
Postgraduate — Biomedical Instrumentation
University of Boumerdes, Algeria
Engineer — Electronics, Communication & Telecommunications

LANGUAGES

French	<i>Native / Fluent</i>
English	<i>Professional</i>
Arabic	<i>Native / Fluent</i>

Automotive (Valeo, Arkamys)
Energy / Smart Grid (Enedis — Linky)
EV Charging / V2G (E2-CAD)
IoT / M2M (Vertical M2M)
Payment (Cartadis, Photomaton, BCM)

VOLUNTEER WORK

Workshop-DIY

Founder — Chelles, France — since 2020

Robotics & STEM workshops for youth (ages 6-12+)

ESP32 robot kit design (WiFi, BLE, servos, sensors)

18+ workshops: Arduino, micro:bit, AI, 3D printing

workshop-diy.org

References available upon request — Eligible for Luxembourg cross-border worker permit